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Predicting Sprint Performance From the Vertical and Horizontal Jumps in National Football League Combine Athletes

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Abstract

Agar-Newman, DJ, MacRae, F, Tsai, M-C, and Klimstra, M. Predicting sprint performance from the vertical and horizontal jumps in National Football League Combine athletes. *J Strength Cond Res* 38(8): 1433–1439, 2024—Identifying fast athletes is an important part of the National Football League (NFL) Combine. However, not all athletes partake in the 36.58-m sprint, and relying on this single test may miss potentially fast athletes. Therefore, the purpose of this study was to determine whether sprinting times can be predicted using simple anthropometric and jumping measures. Data from the NFL Combine between the years 1999–2020 inclusive were used (n = 4,149). Subjects had a mean ($\pm SD$) height = 1.87 ± 0.07 m and body mass = 111.96 ± 20.78 kg. The cross-validation technique was used, partitioning the data into a training set (n = 2,071) to develop regression models to predict time over the 9.14-, 9.14- to 18.29-, 18.29- to 36.58-m, and 36.58-m segments using vertical jump, broad jump, height, and mass as the independent variables. The models were then evaluated against a test set (n = 2,070) for agreement. Statistically significant ($p < 0.01$) models were determined for 9.14-m time (adjusted R^2 = 0.76, *SEE* = 0.05 seconds), 9.14- to 18.29-m time (adjusted R^2 = 0.74, *SEE* = 0.04 seconds), 18.29- to 36.59-m time (adjusted R^2 = 0.79, *SEE* = 0.07 seconds), and 36.58-m time (adjusted R^2 = 0.84, *SEE* = 0.12 seconds). When evaluated against the test set, the models showed biases of −0.05, −0.04, −0.02, and −0.02 seconds and root-mean-square error of 0.07, 0.05, 0.07, and 0.12 seconds for the 9.14-, 9.14- to 18.29-, 18.29- to 36.58-m, and 36.58-m segments, respectively. However, 5–6% of the predictions lay outside of the limits of agreement. This study provides 4 formulae that can be used to predict sprint performance when the 36.58-m sprint test is not performed, and practitioners can use these equations to determine training areas of opportunity when working with athletes preparing for the NFL Combine.

Key Words: max velocity, acceleration, standing long jump, countermovement jump

Introduction

Strength, power, and speed are of paramount importance to success in American football. However, the physiological demands vary across positions with sprinting speed and vertical jump (VJ) particularly important for wide receivers, running backs (RB), and defensive backs (2,11,25). Players in these positions cover the most ground, spend the most time sprinting, and spend more time in the air than players of other positions (2,11,25). In addition, acceleration ability may be important for linemen who must perform repeated short sprints on each play. For players of these positions, acceleration ability may be more important than top speed ability. Therefore, coaches, scouts, and media devote significant attention to a prospective player's sprint performance in the National Football League (NFL) Draft Combine.

The NFL Combine is a test battery designed to inform athlete selection. The Combine tests consist of the 36.58-m (40-yard) sprint with split times at 9.14 m (10-yard) and 18.29 m (20-yard), a 102.27-kg (225-lb) maximum repetition bench press, VJ, standing broad jump (BJ), pro-agility shuttle test, and a 3-cone drill. Approximately 300 of the top draft prospects are invited each year to participate. Although teams base draft selection primarily on competitive playing performance, NFL Combine results are a supplementary factor in the decision-making process (8). This stems from predictive links between Combine performance tests and NFL athlete success in some positions; e.g., sprint performance in the RB position is a valid predictor of an athlete's success in the NFL (14). Therefore, many players spend significant time and resources in the off-season to prepare for the NFL Combine to increase their draft stock. Understanding the relationships between the various NFL Combine tests would better allow practitioners in preparing athletes by identifying potential training areas of opportunity.

In addition, understanding the relationships between the NFL Combine tests may aid teams in estimating sprint performance for those who do not perform a 36.58-m test. Furthermore, there is also the possibility that measurement limitations may mask an athlete's potential sprinting performance. For instance, the 36.58-m timer is begun with a hand-trigger start from a three-point position in the NFL Combine, which makes comparisons to sprints that include reaction time or other starting procedures (e.g., two-point start) difficult (9). In addition, some athletes may have the physical capacity to achieve fast sprint times but may be limited by improper or inefficient technique, whereas simple jump tests (i.e., vertical, or horizontal jumps) are likely less limited by technical ability. Therefore, understanding the relationship between jumping performance and sprinting performance could potentially assist teams to make more educated decisions on draft day.

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Several studies have examined the associations and relationships between jumping ability and sprint performance in a variety of sports; however, these studies often involve soccer or rugby athletes (1,6,13,16,17). Moderate correlations between VJ performance, change of direction without the ball, and sprint performance have been observed in male and female soccer players (13,17). Unfortunately, most studies that claim to investigate the relationship between jumping performance and sprinting ability in NFL Draft eligible players only show the associations between variables (20) and not the relationship, which shows how a change in an independent variable will potentially impact a dependent variable. One of the few studies to report relationships between NFL Combine measures only used data from the 2003 and 2008 combines was published as an abstract (3) and is likely out of date. As sprinting performance is likely important for certain NFL positions, the purpose of this study is to determine whether sprinting times can be predicted using simple anthropometric and jumping measures in NFL Draft eligible players. Based on the previous literature, we hypothesized that sprinting time can be computed from VJ, BJ, and simple anthropometric measures. Therefore, the null hypothesis (Ho) was there will be no relationship between sprinting time and predicted sprinting time, and our alternative hypothesis (Ha) was there will be a relationship between sprinting time and predicted sprinting time.

Methods

Experimental Approach to the Problem

To determine whether sprinting time can be predicted from jumping metrics and simple anthropometric measures, publicly available NFL Combine testing data were retrieved from http://draftscout.com/. The cross-validation technique was used, with the data partitioned into a training set to develop a multiple linear regression model to predict sprint time with VJ, BJ, and basic anthropometric measures (height and mass) as the independent variables and sprint time over the 9.14-, 9.14- to 18.29-, 18.29- to 36.58-m, and 36.58-m segments as the dependent variables. The linear models developed were then used with the test set data to assess the agreement of the calculated times to the actual times.

Subjects

Data used for this study included players who partook in the NFL Combine between the years 1999–2020 inclusive. Data were collected from http://draftscout.com/and have been used in other studies investigating combine performance and deemed accurate (20,21). Data for 7,171 subjects were downloaded, and the study was limited to the subjects who had complete records for the 36.58-m sprint, height, body mass, VJ test, and BJ test, $n = 4{,}149$. Subjects had a mean ($\pm SD$) height = 1.87 ± 0.07 m and body mass = 111.96 ± 20.78 kg, and further descriptive statistics can be found in Table 1. In accordance with the Tri-Council Policy Statement on Ethical Conduct for Research Involving Humans, human research ethics board approval was unnecessary because of the retrospective nature of the study, the availability of the information in the public domain, and the anonymous nature of the results.

Procedures

The procedures for collecting data at the NFL Combine have previously been published (20). For brevity purposes, the following NFL Combine tests were used for the analysis.

Table 1

Descriptive statistics for National Football League Combine athletes.*

	Mean (±*SD*)
Height (m)	1.87 ± 0.07
Body mass (kg)	111.96 ± 20.78
Vertical jump (m)	0.83 ± 0.11
Broad jump (m)	2.88 ± 0.24
9.14-m time (s)	1.68 ± 0.11
18.28- to 36.58-m time (s)	2.02 ± 0.15
36.58-m time (s)	4.82 ± 0.31

*Data for all subjects ($n = 4{,}141$).

Vertical Jump. Subjects stand with their dominant side against the Vertec and extend their arm upward as far as possible. The Vertec is set with the bottom vane either 0.46 or 0.61 m above the subjects' outstretched arm. With both feet firmly planted to the ground, the subjects may use a countermovement and arm-swing, jumping as high as possible attempting to reach the highest vane on the Vertec from the floor. Any shuffling of the feet before takeoff results in a failed attempt, and the jump did not count. After the first jump, the subject then resets their feet and has a second attempt to touch a higher vane. The distance from the subjects' outstretched hand at the start of the movement to the highest vane touched of the 2 attempts is taken as their jump height and measured in 0.01-m increments.

Broad Jump. Subjects start with their toes behind the start line. Using a countermovement and arm swing, subjects jump as far as possible in the horizontal direction. Subjects then must land balanced with both feet planted, although falling forward is allowed; however, falling backward disqualifies the jump. Jump distance is measured from the start line to the heel of the nearest foot to the start line. Jumps are measured to the nearest 0.03-m increment. Subjects receive 2 attempts, and the farther jump recorded is taken for analysis.

36.58-m Sprint. Timing gates are placed at the 9.14-, 18.29-, and 36.58-m marks. The subject holds a 3-point stance for 2 seconds and receives the command "you can go" from the tester. The subject then runs 36.58 m as fast as possible. The timer is started when the subject's down-hand separates from the turf. After running 36.58 m, the subject jogs back to the starting area and takes approximately 10–15 minutes before commencing their second attempt. Rolling starts are not allowed.

Statistical Analyses

Data from 7,174 athletes were downloaded, and incomplete sets of data (missing any independent or dependent variables) were excluded from the analysis, resulting in 4,071 complete sets of data. The analysis consisted of 3 parts. First, the data were randomized by position and testing date and divided into a training set ($n = 2{,}071$) and test set ($n = 2{,}070$) using Microsoft Excel (Version 2209, Redmond, WA). Second, backward stepwise multiple regressions were conducted to determine the relationship between 9.14-m time, 9.14- to 18.29-m time, 18.29- to 36.58-m time, 36.58-m time, and the jumping and anthropometric measures using the training set of data (IBM SPSS Statistics, Version 27.0.1.0, Armonk, NY), and alpha was set to 0.05. Finally, Bland and Altman plots were used to assess the level of agreement between measured sprinting times and calculated sprinting times

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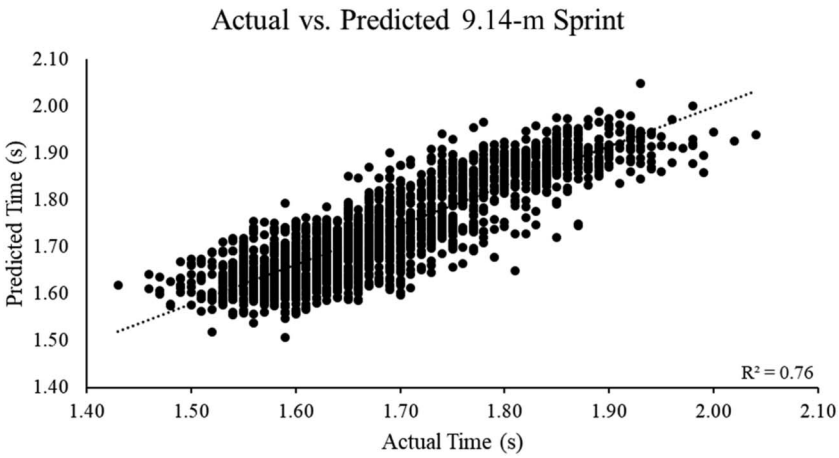

Figure 1. Predicted sprint times using jumping measures and basic anthropometrics using equation 1.

using the test set of data. A range of agreement was defined as mean bias ±2 *SD* with 95% of values within the limits (5). The limit of agreement of difference was defined a priori as 5% of sprinting time, which is clinically acceptable. The goodness-of-model fit was assessed using root-mean-square error.

Results

When predicting 9.14-m time, there were significant standardized coefficients for mass ($\beta = 0.50$), VJ ($\beta = -0.09$), and BJ ($\beta = -0.37$). For 9.14- to 18.29-m time, there were significant standardized coefficients for height ($\beta = 0.06$), mass ($\beta = 0.48$), VJ ($\beta = -0.11$), and BJ ($\beta = -0.31$). There were also significant standardized coefficients for 18.29- to 36.58-m time for mass ($\beta = 0.53$), VJ ($\beta = -0.10$), and BJ ($\beta = -0.34$). All previously noted coefficients were significant at the $p < 0.01$ level. For 36.58-m time, there were significant standardized coefficients for height ($\beta = 0.03$, $p = 0.01$), mass ($\beta = 0.52$), VJ ($\beta = -0.10$), and BJ ($\beta = -0.36$) with mass, VJ, and BJ significant at the $p < 0.01$ level. The equations for predicting 9.14-m time (1), 9.14- to 18.29-m time (2), 18.29- to 36.58-m time (3), and 36.58-m time (4) are presented below:

$$9.14 - m\,time = 1.941 + 0.003 \bullet \text{mass} - 0.90 \bullet \text{VJ} - 0.164 \bullet \text{BJ} \quad (1),$$

$$9.14 - 18.29 - m\,time = 1.15 + 0.06 \bullet \text{height} + 0.002 \bullet \text{mass} - 0.073 \bullet \text{VJ} - 0.093 \bullet \text{BJ} \quad (2),$$

$$18.29 - 36.58 - m\,time = 2.249 + 0.004 \bullet \text{mass} - 0.188 \bullet \text{VJ} - 0.231 \bullet \text{BJ} \quad (3),$$

$$36.58 - m\,time = 5.284 + 0.148 \bullet \text{height} + 0.008 \bullet \text{mass} - 0.300 \bullet \text{VJ} - 0.471 \bullet \text{BJ} \quad (4).$$

The adjusted R^2 and *SEE* for equation 1 were 0.76 and 0.05 seconds, for equation 2 were 0.74 and 0.04 seconds, for equation 3 were 0.79 and 0.07 seconds, and for equation 4 were 0.84 and 0.12 seconds, respectively. These equations were used to estimate 9.14-, 9.14- to 18.29-, 18.29- to 36.58-, and 36.58-m times using the test set of data (Figures 1–4). The goodness-of-the-model fit was assessed in Figures 5–8 showed biases of −0.05, −0.04, −0.02, and −0.02 seconds and a root-mean-square error of 0.07, 0.05, 0.07, and 0.12 seconds, respectively. However, 5.75% of

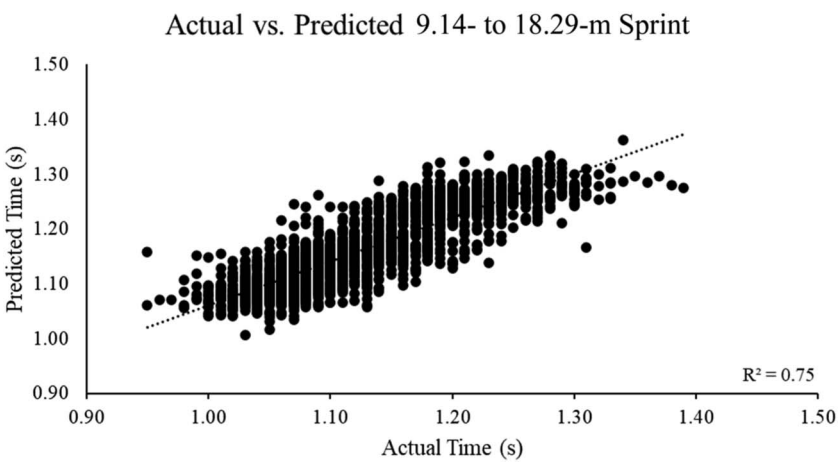

Figure 2. Predicted sprint times using jumping measures and basic anthropometrics using equation 2.

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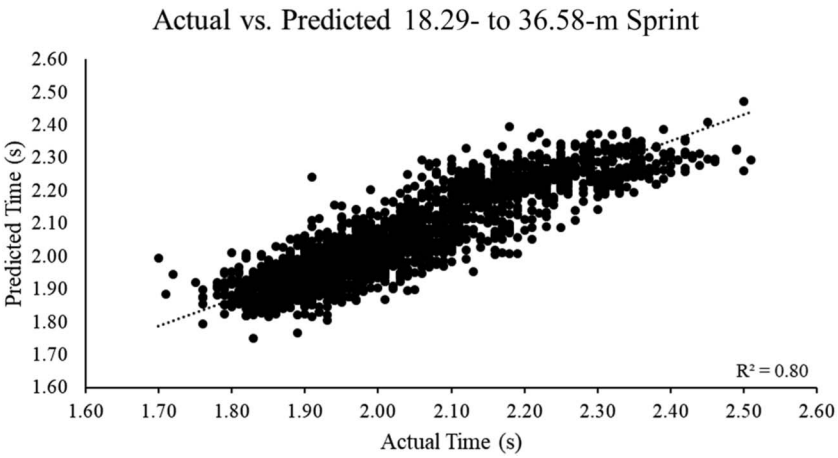

Figure 3. Predicted sprint times using jumping measures and basic anthropometrics using equation 3.

observations in the 9.14-m sprint, 5.46% of observations for the 9.14- to 18.29-m sprint, 5.65% of observations in the 18.29- to 36.58-m sprint, and 5.02% of observations in the 36.58-m sprint fell outside the limits of agreement.

Discussion

This study revealed that there is a relationship between sprint time for 9.14-, 9.14- to 18.29-, 18.29- to 36.58-, and 36.58-m sprints and predicted sprint time from VJ, BJ, and basic anthropometrical measurements in elite American football players. The proposed formulae provide coaches with tools to predict sprint performance from incomplete sets of combined data. Furthermore, these formulae provide coaches with tools to compare an athlete's actual sprint performance against the theoretical time, possibly revealing technical deficiencies or areas of opportunity for training. In addition, the unstandardized beta values in these formulae can be used to show how an improvement in a specific component (i.e., independent variable) will impact sprint time. However, coaches should be cautious with the precision of these models as >5% of the data fell outside of the limits of agreement.

Little research has been conducted to establish predictive validity between the different tests in NFL Combine athletes. A similar analysis to this study was conducted in 2011; however, they only included data from the 2003 and 2008 combines, resulting in a much smaller sample size (n = 443) and failed to test the accuracy of the model (3). This study used a larger sample (n = 4,071) and a regression analysis aimed to establish predictive validity of BJ, VJ, height, and mass for 9.14-, 9.14- to 18.29-, 18.29- to 36.58-, and 36.58-m sprint times. In addition, this study tests the agreement of the modeled time against the actual time using a test set of data. This study elaborates on the findings from existing literature and goes further to provide equations for the calculation of sprint time from jumping metrics and tests the accuracy of the prediction.

The literature supports our findings that VJ, BJ, height, and body mass are closely related to sprint performance in athletes (3,13,15,17). As expected, increased mass results in slower sprint times in this population of athletes. However, an interesting finding was that increased height also resulted in slower sprint times for the 9.14- to 18.29-m segment and 36.58-m sprint. Although increased height likely has a positive impact on sprinting performance through larger contact lengths allowing taller athletes a greater distance/time to produce force, taller athletes in this population are likely increasing their body mass disproportionately, offsetting the advantage the increased contact length

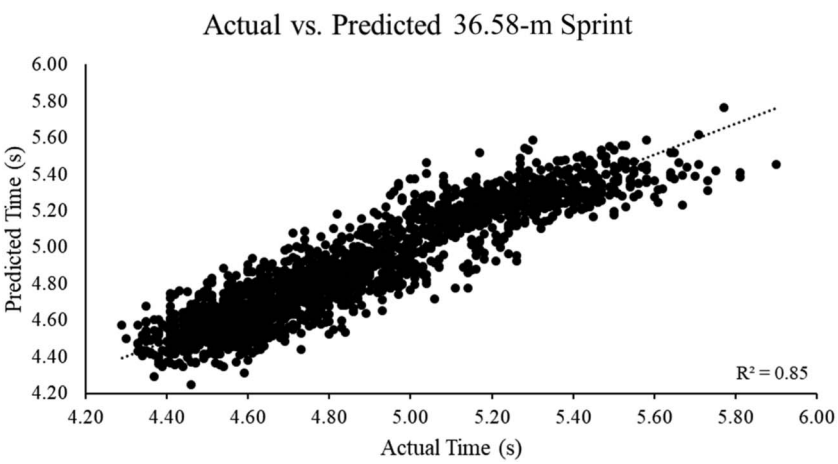

Figure 4. Predicted sprint times using jumping measures and basic anthropometrics using equation 4.

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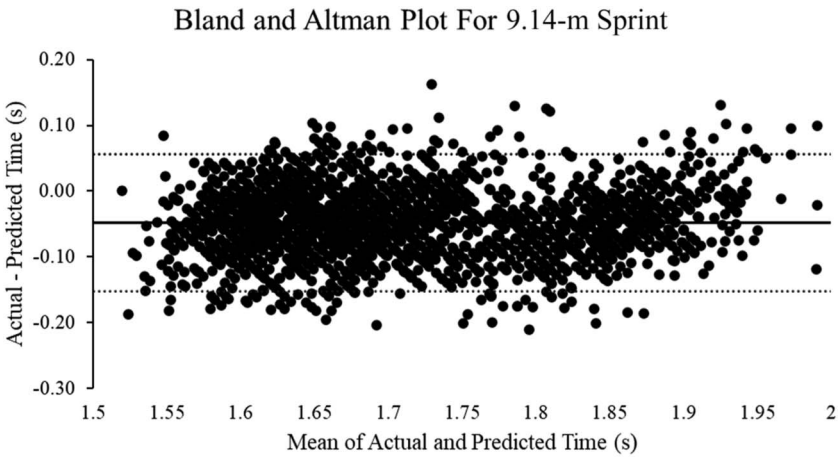

Figure 5. Bland and Altman plot showing the bias (solid line) and limits of agreement (dashed lines) for equation 1.

provides to maximize their momentum into collisions (26). The relationship between VJ, BJ, and sprint time may be due to the importance of the producing mass-specific force in the horizontal direction during acceleration (19), such as the BJ, and vertical mass-specific force during maximal velocity, such as the VJ (27). Interestingly, the standardized coefficients for the BJ were larger than the VJ for all sprint distances. This difference can possibly be attributed to the fact that the force vector during a BJ has both a vertical and horizontal component such as sprinting (7). This means that BJ performance is not only dependent on the athlete's ability to produce mass-specific vertical forces, which determines flight time during the jump but also the athlete's ability to produce large horizontal forces, which determines their horizontal velocity at take-off (7). Possibly, a stronger argument supporting the relationship between jumping and sprinting ability is due to both tasks requiring similar underlying physical qualities such as high levels of thigh muscle cross-sectional area, optimal body composition, and a high percentage of type II fibers (12,18). In addition, an optimal level of mobility in the hips and ankles likely impacts the landing position of the athlete during the BJ, which could be an important factor to sprinting performance, but further research would need to confirm this.

These equations provide coaches with a tool to assess an athlete's sprinting ability against their predicted sprinting ability through jump performance and anthropometrics. As with all predictive equations, there is inherently some error; therefore, coaches should view the difference between the predicted and actual time as an item that warrants further investigation, with larger differences (outside of the root-mean-square error) warranting greater attention. This allows coaches to identify potential areas of opportunity. Simply said if the athlete is running faster than their predicted time, coaches may look to examine the athlete's underlying physical constraints (i.e., force expression, etc), and if the athlete is running slower than their predicted time, coaches may want to examine the athlete's technical constraints (i.e., direction and timing of force application in the sprint, etc). Some examples of technical aspects that coaches may want to examine include the position of the center of mass during the stance phase because elite sprinters have greater horizontal orientation of the center of mass during the initiation of the sprint, allowing them to better direct force in the horizontal direction when compared with well-trained sprinters in a 10-m sprint (23). Some examples of physical aspects include the absolute and relative cross-sectional area of the psoas major and gluteus maximus

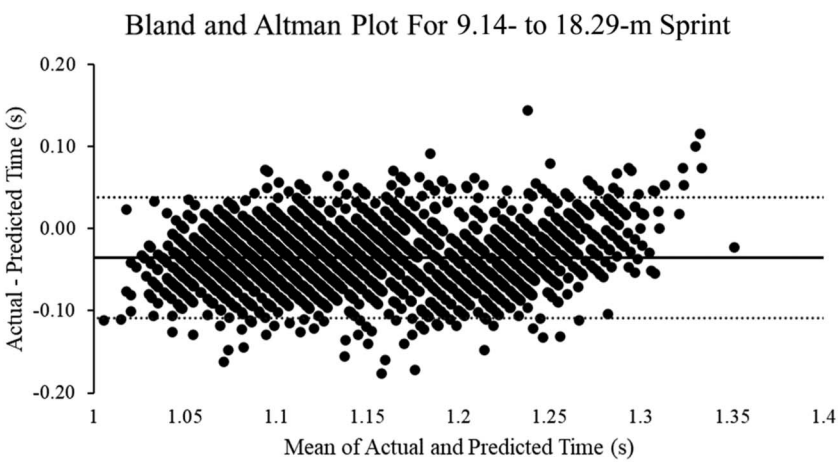

Figure 6. Bland and Altman plot showing the bias (solid line) and limits of agreement (dashed lines) for equation 2.

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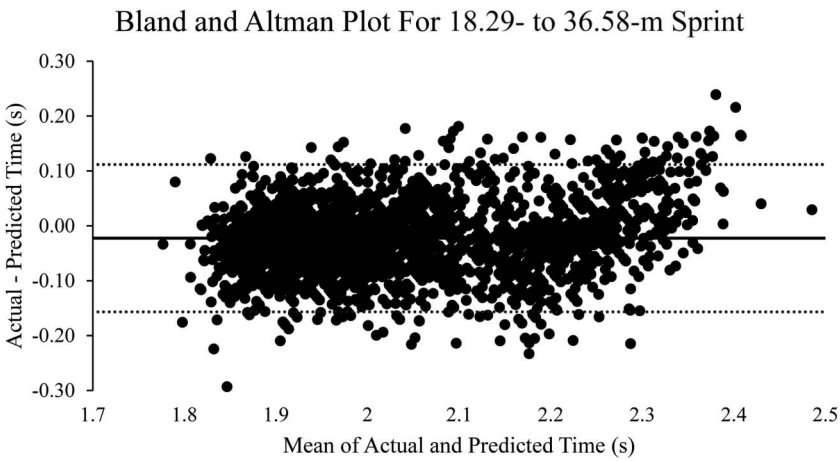

Figure 7. Bland and Altman plot showing the bias (solid line) and limits of agreement (dashed lines) for equation 3.

muscles, with sprinters having larger areas when compared with nonsprinters (24) and have likely performed more resistance training (23). Furthermore, inadequate development of specific musculature may lead to poor performance in the sprinting assessment. Also, the specific muscles required for a VJ and BJ differ, with the VJ relying more heavily on muscles about the knee compared with the BJ (22), and the BJ requiring the athletes to transition rapidly from a powerful hip extension for take-off to flexion for the landing and flexibility requirements around the hips and ankles to maximize the landing. Potential imbalances in VJ, BJ, and sprint could possibly indicate a relative area of opportunity in one of these aforementioned areas. As such, it may be prudent to explore the underlying causes of the variation in performance in the NFL Combine movements as a potential explanation for a poor performance in 1 area. It is plausible that this could again be due to a technical deficiency. Furthermore, the unstandardized beta values in the equations could be used to estimate how much an improvement in a specific component will impact the sprint time. This would be performed by multiplying the beta value by the athlete's current value and then doing the same with an improved value and comparing the difference, which would signify the potential improvement in seconds.

Although this study has provided 4 equations that can be used to predict sprint performance from the predictors with high accuracy, there are several limitations. First, the protocols used in the NFL Combine are not perfect because the sprint test timer is started with a hand-trigger and a three-point start. Although this may increase the face validity for linemen who start in a three-point stance, it reduces the validity of this test for skilled players and potentially the reliability of the test (10). Furthermore, the VJ and BJ tests both allow for the athlete to perform a countermovement and allow the upper body to contribute to the force production through using an arm swing. Although this possibly makes the VJ and BJ a more global test of upper- and lower-body force production, coordination, and more ecologically valid, these tests fail to isolate the lower-body force-producing abilities (e.g., jumping tasks performed by akimbo), which may be of interest to practitioners. In addition, the calculation of VJ height using a Vertec likely inflates jump height compared with calculation methods using force plates using the impulse-momentum method (4). This is due to the athlete's ability to gain extra height through elevating the reaching arm's shoulder and dropping the opposing shoulder, despite the center of mass not traveling higher and discrepancies with how standing reach is measured, which

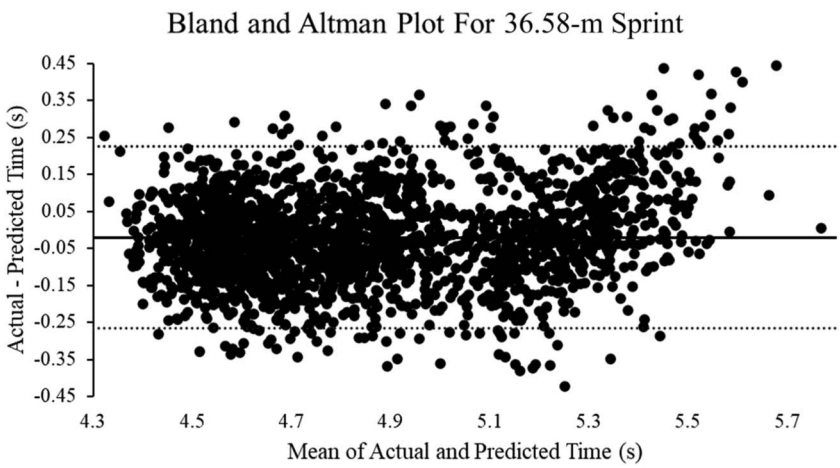

Figure 8. Bland and Altman plot showing the bias (solid line) and limits of agreement (dashed lines) for equation 4.

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may impact the ability to generalize these equations to VJ height measured from force plates (4). Finally, this data set comprises athletes with high levels of performance across the tests used, although practitioners should feel confident interpolating results within the range of times used in this study, caution should be used when extrapolating the results to populations outside of these performance levels. Therefore, further research should be conducted to validate these equations with more general populations.

Practical Applications

This study provides 4 formulae that can be used to predict sprint performance using height, body mass, VJ, and BJ. This will allow practitioners to predict sprint time with incomplete sets of data and to identify athletes who may have possibly underperformed or overperformed in the 36.58-m sprint. Therefore, not every slow or fast athlete is equivalent, and those with superior jumping skills may still be draft candidates. Furthermore, practitioners can use these equations when performing a needs analysis to identify an athlete's strengths and weaknesses to improve sprinting performance.

References

1. Agar-Newman DJ, Klimstra MD. Efficacy of horizontal jumping tasks as a method for talent identification of female rugby players. *J Strength Cond Res* 29: 737–743, 2015.
2. Edwards T, Spiteri T, Piggott B, Haff GG, Joyce C. A narrative review of the physical demands and injury incidence in American football: Application of current knowledge and practices in workload management. *Sports Med* 48: 45–55, 2018.
3. Fairchild BP, Amonette WE, Spiering BA. Prediction models of speed and agility in NFL Combine attendees. *J Strength Cond Res* 25: S96, 2011.
4. Ferreira LC, Schilling BK, Weiss LW, Fry AC, Chiu LZ. Reach height and jump displacement: Implications for standardization of reach determination. *J Strength Cond Res* 24: 1596–1601, 2010.
5. Giavarina D. Understanding Bland Altman analysis. *Biochem Med* 25: 141–151, 2015.
6. Gunaydin G. Is there a relationship between horizontal jump and sprint performance in professional amputee football players? *Ann Med Res* 26: 2454–2458, 2019.
7. Harry JR, Krzyszkowski J, Chowning LD, Kipp K. Phase-specific force and time predictors of standing long jump distance. *J Appl Biomech* 37: 400–407, 2021.
8. Hartman M. Competitive performance compared to combine performance as a valid predictor of NFL draft status. *J Strength Cond Res* 25: S105–S106, 2011.
9. Haugen T, Buchheit M. Sprint running performance monitoring: Methodological and practical considerations. *Sports Med* 46: 641–656, 2016.
10. Haugen TA, Tønnessen E, Seiler SK. The difference is in the start: Impact of timing and start procedure on sprint running performance. *J Strength Cond Res* 26: 473–479, 2012.
11. Hoffman JR. The applied physiology of American football. *Int J Sports Physiol Perform* 3: 387–392, 2008.
12. Hopwood HJ, Bellinger PM, Compton HR, Bourne MN, Minahan C. The relevance of muscle fiber type to physical characteristics and performance in team-sport athletes. *Int J Sports Physiol Perform* 18: 223–230, 2023.
13. Köklü Y, Alemdaroğlu U, Özkan A, Koz M, Ersöz G. The relationship between sprint ability, agility and vertical jump performance in young soccer players. *Sci Sports* 30: e1–e5, 2015.
14. Kuzmits FE, Adams AJ. The NFL combine: Does it predict performance in the National Football League? *J Strength Cond Res* 22: 1721–1727, 2008.
15. López-Segovia M, Marques M, van den Tillaar R, González-Badillo J. Relationships between vertical jump and full squat power outputs with sprint times in U21 soccer players. *J Hum Kinet* 30: 135–144, 2011.
16. Loturco I, Pereira LA, Fílter A, et al. Curve sprinting in soccer: Relationship with linear sprints and vertical jump performance. *Biol Sport* 37: 277–283, 2020.
17. Merino-Munoz P, Vidal-Maturana F, Aedo-Munoz E, Villaseca-Vicuna R, Perez-Contreras J. Relationship between vertical jump, linear sprint and change of direction in Chilean female soccer players. *J Phys Educ Sport* 21: 2737–2744, 2021.
18. Methenitis SK, Zaras ND, Spengos KM, et al. Role of muscle morphology in jumping, sprinting, and throwing performance in participants with different power training duration experience. *J Strength Cond Res* 30: 807–817, 2016.
19. Morin JB, Edouard P, Samozino P. Technical ability of force application as a determinant factor of sprint performance. *Med Sci Sports Exerc* 43: 1680–1688, 2011.
20. Robbins DW. Relationships between National Football League combine performance measures. *J Strength Cond Res* 26: 226–231, 2012.
21. Robbins DW. The normalization of explosive functional movements in a diverse population of elite American football players. *J Strength Cond Res* 26: 995–1000, 2012.
22. Robertson DGE, Fleming D. Kinetics of standing broad and vertical jumping. *Can J Sports Sci* 12: 19–23, 1987.
23. Slawinski J, Bonnefoy A, Levêque J-M, et al. Kinematic and kinetic comparisons of elite and well-trained sprinters during sprint start. *J Strength Cond Res* 24: 896–905, 2010.
24. Tottori N, Suga T, Miyake Y, et al. Trunk and lower limb muscularity in sprinters: What are the specific muscles for superior sprint performance? *BMC Res Notes* 14: 74, 2021.
25. Wellman AD, Coad SC, Goulet GC, McLellan CP. Quantification of accelerometer derived impacts associated with competitive games in National Collegiate Athletic Association Division I college football players. *J Strength Cond Res* 31: 330–338, 2017.
26. Weyand PG, Sandell RF, Prime DNL, Bundle MW. The biological limits to running speed are imposed from the ground up. *J Appl Physiol* 108: 950–961, 2010.
27. Weyand PG, Sternlight DB, Bellizzi MJ, Wright S. Faster top running speeds are achieved with greater ground forces not more rapid leg movements. *J Appl Physiol* 89: 1991–1999, 2000.

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